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BONDED SHOWERHEAD ASSEMBLY AND METHOD FOR USING THE SAME

Abstract

The present disclosure generally relate to a processing chamber comprising a showerhead assembly. The showerhead assembly comprises a heat transfer plate having a first side and a second side, the heat transfer plate having cooling channels formed adjacent the first side of the heat transfer plate and a heater formed in the second side of the heat transfer plate, a showerhead plate having a first side coupled to the second side the heat transfer plate, a first gas passageway extending through the heat transfer plate to the showerhead plate, and a gas distribution plate (GDP) bonded to a second side of the showerhead plate, the GDP having a first plurality of holes formed therethrough, first ends of the first plurality of holes open to the showerhead plate. The showerhead plate comprises a metal backing and a plurality of anodized gas paths extending within the metal backing.

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Background/Summary

BACKGROUND

Field

[0001] Embodiments of the present disclosure generally relate to showerhead assembly, a processing chamber having a showerhead assembly, and a method for using the showerhead assembly.

Description of the Related Art

[0002] Semiconductor substrates are processed for a wide variety of applications, including the fabrication of integrated devices and micro devices. One such processing device is an etch processing chamber. During processing, the substrate is positioned on a substrate support within the etch processing chamber. Gas is introduced into the etch chamber via a showerhead assembly and ignited into a plasma for etching a substrate.

[0003] However, repeated exposure of the gas can cause components of the showerhead assembly to degrade or corrode overtime. When components of the showerhead assembly breakdown, the processing chamber may no longer be functional until the corroded components can be replaced with new components. If the corroded components are unable to be replaced due to the complexity of the showerhead assembly, the processing chamber may be unusable.

[0004] Therefore, a need exists for improved showerhead assembly.

SUMMARY

[0005] The present disclosure generally relates to showerhead assembly, a processing chamber having the showerhead assembly, and a method for using the showerhead assembly. In one example, a showerhead assembly includes a heat transfer plate having a first side and a second side, the heat transfer plate having cooling channels formed adjacent the first side of the heat transfer plate and a heater formed in the second side of the heat transfer plate, a showerhead plate having a first side coupled to the second side the heat transfer plate, a first gas passageway extending through the heat transfer plate to the showerhead plate, and a gas distribution plate (GDP) bonded to a second side of the showerhead plate, the GSP having a first plurality of holes formed therethrough, first ends of the first plurality of holes open to the showerhead plate.

[0006] In another example, a showerhead assembly includes a heat transfer plate having a first side coupled to a second side by a sidewall, the heat transfer plate having cooling channels formed adjacent the first side of the heat transfer plate and a heater disposed in a heater receiving channel formed in the second side of the heat transfer plate, a showerhead plate having a first side coupled to the second side the heat transfer plate, and a gas distribution plate bonded to a second side of the showerhead plate, the gas distribution plate having a first plurality of holes formed therethrough, first ends of the first plurality of holes open to the showerhead plate.

[0007] In another example, a showerhead assembly includes a heat transfer plate having a first side coupled to a second side by a sidewall, the heat transfer plate having cooling channels formed adjacent the first side of the heat transfer plate and a heater disposed in a heater receiving channel formed in the second side of the heat transfer plate, a showerhead plate having a first side coupled to the second side the heat transfer plate, wherein the showerhead plate comprises a plurality of anodized gas channels in a second side of the showerhead plate, a first gas passageway extending through the heat transfer plate to the plurality of anodized gas channels of the showerhead plate, and a gas distribution plate bonded to a second side of the showerhead plate, the gas distribution plate having a first plurality of holes formed therethrough, first ends of the first plurality of holes open to the plurality of anodized gas channels formed in the showerhead plate.

[0008] In yet another example, a processing chamber is provided that includes a chamber body having an internal volume, a substrate support disposed in the internal volume, a lid disposed on the chamber body and enclosing the internal volume, a heat transfer plate having a first side coupled to a second side by a sidewall, the heat transfer plate having cooling channels formed adjacent the first side of the heat transfer plate and a heater disposed in a heater receiving channel formed in the second side of the heat transfer plate, a showerhead plate having a first side coupled to the second side the heat transfer plate, wherein the showerhead plate comprises a plurality of gas channels in a second side of the showerhead plate, and a gas distribution plate bonded to a second side of the showerhead plate, the gas distribution plate having a first plurality of holes formed therethrough, first ends of the first plurality of holes open to the plurality of gas channels formed in the showerhead plate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope, may admit to other equally effective embodiments.

[0010] FIG. **1** is a schematic cross-sectional view of a processing chamber according to one or more embodiments of the disclosure.

[0011] FIG. **2** illustrates a conventional showerhead assembly.

[0012] FIG. **3**A illustrates a showerhead assembly, according to one embodiment.

[0013] FIG. **3**B illustrates a showerhead plate of the showerhead assembly of FIG. **3**A.

[0014] FIG. **3**C illustrates anodized components of the showerhead assembly of FIG. **3**A.

[0015] FIG. **4**A illustrates a bottom view of the showerhead plate of the showerhead assembly of FIGS. **3**A-**3**C comprising two zones, according to one embodiment.

[0016] FIG. **4**B illustrates a bottom view of the showerhead plate of the

[0017] showerhead assembly of FIGS. **3**A-**3**C comprising three zones, according to another embodiment.

[0018] FIG. **5** illustrates the pressure sensor and pressure sensing channel of the showerhead assembly of FIGS. **3**A-**3**C, according to one embodiment.

[0019] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

[0020] The present disclosure generally relates to showerhead assembly, a processing chamber having a showerhead assembly, and a method for using the showerhead assembly. The showerhead assembly includes a heat transfer plate having a first side and a second side, the heat transfer plate having cooling channels formed adjacent the first side of the heat transfer plate and a heater formed in the second side of the heat transfer plate, a showerhead plate having a first side coupled to the second side the heat transfer plate, a first gas passageway extending through the heat transfer plate to the showerhead plate, and a gas distribution plate (GDP) bonded to a second side of the showerhead plate, the GDP having a first plurality of holes formed therethrough, first ends of the first plurality of holes open to the showerhead plate.

[0021] Due to the showerhead plate **128** and the GDP **302** being coupled together by the adhesive and being distinct from the heat transfer plate **130** (i.e., not embedded within the heat transfer plate **130** like a conventional showerhead assembly), the showerhead plate **128** and the GDP **302** can be easily replaced when needed. Moreover, the showerhead plate **128** can be anodized, and the thin GDP **302** may have mechanically drilled holes, thus resulting in reduced costs when compared to a conventional showerhead assembly.

[0022] FIG. **1** is a schematic cross-sectional view of a processing chamber **100** according to one or more embodiments of the disclosure. The exemplary processing chamber **100** is suitable for patterning a material layer disposed on a substrate **116** in the processing chamber **100**. The exemplary processing chamber **100** is suitable for performing a patterning process. The processing chamber **100** may be a plasma etch chamber, a plasma enhanced chemical vapor deposition chamber, a physical vapor deposition chamber, a plasma treatment chamber, an ion implantation chamber, or other suitable vacuum processing chamber.

[0023] The processing chamber **100** has a body **140**. The body **140** generally has four external surfaces. The body **140** includes a source block **102**, a process block **104**, a flow block **106**, and an exhaust block **108**. It should be appreciated that the blocks may be one or more combination of blocks. For example, the exhaust block **108** is integral with and a part of the flow block **106** and made as a single unified body. The flow block **106** is part of a pumping port assembly **111** that includes a substrate support chassis **154**. The source block **102**, the process block **104** and the flow block **106** collectively enclose a process region **112**. During operation, a substrate **116** may be positioned on a substrate support assembly **118** and exposed to a process environment, such as plasma generated in the process region **112**. Exemplary process which may be performed in the processing chamber **100** may include etching, chemical vapor deposition, physical vapor deposition, implantation, plasma annealing, plasma treating, abatement, or other plasma processes. Vacuum may be maintained in the process region **112** by suction from an exhaust port **181** formed in the exhaust block **108** through one or more evacuation channels, i.e., evacuation channels **114**, defined in the flow block **106**.

[0024] The process region **112** and the evacuation channels **114** are substantially symmetrically about a central axis **110** to provide symmetrical electrical current, gas flow, thermal and pressure uniformity to establish uniform process conditions.

[0025] The source block **102** includes a showerhead assembly **120** (also referred to as an upper electrode or anode) may be isolated from and supported by the process block **104** by an isolator **122**. Alternatively, the source block **102**, may be electrically coupled to the process block **104**, and grounded. A lid **131** is disposed on the isolator **122**. The showerhead assembly **120** may include a showerhead plate **128** attached to a heat transfer plate **130**. The showerhead assembly **120** may be connected to a gas source **132** through a gas inlet tube **126**.

[0026] The gas source **132** may include one or more process gas sources and may additionally include inert gases, non-reactive gases, and reactive gases, if desired. Examples of process gases that may be provided by the gas source **132** include, but are not limited to, hydrocarbon containing gas including methane (CH_4), sulfur hexafluoride (SF_6), silicon chloride ($SiCl_4$), carbon tetrafluoride (CF_4), hydrogen bromide (HBr), hydrocarbon containing gas, argon gas (Ar), chlorine (Cl_2), nitrogen (N_2), helium (He) and oxygen gas (O_2). Additionally, process gasses may include nitrogen, chlorine, fluorine, oxygen and hydrogen containing gases such as BCl_3, C_2F_4, C_4F_8, C_4F_6, CHF_3, CH_2F_2, CH_3F, NF_3, NH_3, CO_2, SO_2, CO, N_2, NO_2, N_2O and H_2 among others.

[0027] The showerhead plate **128**, the heat transfer plate **130**, and the gas inlet tube **126** may be all fabricated from a radio frequency (RF) conductive material, such as aluminum or stainless steel. The showerhead assembly **120** may be coupled to a RF power source **124** via a match circuit **125** and the conductive gas inlet tube **126**. The conductive gas inlet tube **126** may be coaxial with the central axis **110** of the processing chamber **100** so that both RF power and processing gases from the gas source **132** are symmetrically provided.

[0028] The process block **104** is disposed on the flow block **106**. An RF gasket for grounding and an O-ring seal is disposed between the process block **104** and the flow block **106**. Alternately, the process block **104** and flow block **106** are combined and made as a single unified body with no RF gasket for grounding and O-ring seal between them.

[0029] The process block **104** encloses the process region **112**. The process block **104** may be fabricated from a conductive material resistive to processing environments, such as aluminum or stainless steel. The substrate support assembly **118** may be centrally disposed within the process block **104** and positioned to support the substrate **116** in the process region **112** symmetrically about the central axis **110**.

[0030] A slit valve opening **142** may be formed through the process block **104** to allow passages of the substrate **116**. A slit valve **144** may be disposed outside the process block **104** to selectively open and close the slit valve opening **142**.

[0031] The process block **104** is disposed on the flow block **106**. The flow block **106** provides flow paths between the process region **112** defined in the process block **104** and the exhaust block **108**. The flow block **106** also provides an interface between the substrate support assembly **118** and the atmospheric environment exterior to the processing chamber **100**.

[0032] The flow block **106** has through-holes **170** and evacuation channels **114**. The through-holes **170** are maintained at atmospheric pressure and provide access to the substrate support assembly **118**. The evacuation channels **114** are maintained at vacuum and provides a fluid path for removing gasses from the process region **112** to outside the processing chamber **100**.

[0033] FIG. **2** illustrates a conventional showerhead assembly **200** that may be used to replace the showerhead assembly **120** in the processing chamber **100** described above. The conventional showerhead assembly **200** is provided for comparison to the novel showerhead assembly **120** as further detailed below with respect to FIGS. **3**A-**3**C.

[0034] Continuing to refer to FIG. **2**, the showerhead assembly **200** comprises a gas distribution plate (GDP) **202**, a showerhead plate **204** disposed over the GDP **202**, and a heat transfer plate **206** disposed over the showerhead plate **204**. A lid **221** is disposed on the chamber body **140**. In the case of a powered upper electrode, an isolator **211** is disposed between the lid **221** and showerhead assembly **200**, adjacent to the sides of showerhead plate **204** and the heat transfer plate **206**. The heat transfer plate **206** comprises a plurality of coolant channels **218** embedded therein. The showerhead plate **204** has one or more trenches or slots **215** formed therein. One or more heaters **216** and one or more plugs **217** are disposed within the one or more slots **215**. The GDP **202** is tensioned against the showerhead plate **204** using a plurality of screws **212** and one or more springs **208**. The screws **212** are thread into threaded inserts **222** inserted into the GDP **202**. Thus, the GDP **202** must be thick enough to accommodate the inserts **222**, which drives up the cost of the GDP **202** while also making the gas holes (not shown) formed through the GDP **202** more costly and time consuming to fabricate. The springs **208** are utilized to maintain good contact between the GDP **202** and showerhead plate **204** over a wide temperature range without having to overly torque the screws **212** and risking damage to the GDP **202**. A plurality of o-rings (not shown) are disposed between the screws **212** and the showerhead plate **204** and/or heat transfer plate **206**. The o-rings prevent gases from flowing out of the chamber and further protect the GDP **202**. In one example, over **20** screws **212** and over **20** o-rings may be utilized.

[0035] A plurality of gas distribution channels **210** are embedded in the showerhead plate **204** above the GDP **202**, wherein the gas distribution channels **210** are spaced from the GDP **202**. The heat transfer plate **206** and the showerhead plate **204** may comprise aluminum, for example. The portion of the showerhead plate **204** disposed between the GDP **202** and the gas distribution channels **210** fails to distribute heat well during operation.

[0036] Because the coolant channels **218** and the gas distribution channels **210** are embedded within the heat transfer plate **206** and the showerhead plate **204**, the coolant channels **218** and the gas distribution channels **210** are capped using diffusion bonding or welding when formed, making the processes available for protecting the interior surfaces of the gas distribution channels **210** from the process gases flowing therein very limited. For example, the atomic layer deposition techniques conventionally used to coat the interior surfaces of the gas distribution channels **210** within the showerhead plate **204** are very expensive and time consuming to apply, which significantly adds to the cost of the showerhead plate **204**. Moreover, the diffusion bonding in and of itself is costly.

[0037] The GDP **202** comprises silicon, for example, and has a thickness in the y-direction of about 10 mm. The GDP **202** has a plurality of holes (not shown) disposed therethrough to distribute gas to a chamber disposed below. Due to the thickness of the GDP **202**, the plurality of holes are typically laser drilled, which is costly and time consuming. Due to the various factors discussed (i.e., the springs, the diffusion bonding and ALD processes and laser drilled holes), the showerhead assembly **200** can be expensive to manufacture.

[0038] FIG. **3**A illustrates a partial sectional view of the showerhead assembly **120**, according to one embodiment. FIG. **3**B illustrates a showerhead plate **128** of the showerhead assembly **120** of FIG. **3**A. FIG. **3**C illustrates anodized components of the showerhead assembly **120** of FIG. **3**A. The showerhead assembly **120** may comprise one or more zones, as discussed in FIGS. **4**A-**4**B.

[0039] The showerhead assembly **120** comprises a showerhead plate **128**, and a heat transfer plate **130** disposed over the showerhead plate **128**. A thermal gasket or other thermal interface material (not shown) may be disposed between the showerhead plate **128** and the heat transfer plate **130**. The showerhead plate **128** comprises a metal backing plate, such as an aluminum plate, and gas channels **310** disposed within the metal backing plate. The showerhead plate **128** is bonded to a gas distribution plate (GDP) **302**, where the GDP **302** is disposed in thermal contact with the showerhead plate **128**. The gas channels **310** formed in the showerhead plate **128** are open to the bottom surface of the showerhead plate **128**, such that the GDP **302** forms the bottom boundary of each gas path **310**, thus eliminating the need to weldingly seal the gas paths as conventionally done while fabricating the showerhead plate **204** of the conventional showerhead assembly **200** illustrated in FIG. **2**. One or more screws or bolts **328** are used to connect the metal backing of the showerhead plate **128** to a heat transfer plate **130**. Because the bolts **328** are connected to the metal backing of the showerhead plate **128**, springs as used in conventional showerhead assemblies **200** are unnecessary, as the bolts **328** are not threaded into the GDP **302**. Since the GDP **302** no longer needs to have sufficient thickness to accommodate threaded inserts (or a treaded hole), the GDP **302** is significantly thinner than the conventional GDP **202**, thus making the GDP **302** significantly less expensive and easier to fabricate.

[0040] The GDP **302** comprises a plurality of holes **360** formed therethrough. Each of the holes **360** has a same size in order to evenly distribute gas within a given zone **342-346** (shown in FIGS. **4**A-**4**B). First ends of the plurality of holes **360** of the GDP **302** are open to the gas channels **310**. The gas channels **310** are disposed directly on the GDP **302**, and may have a serpentine-like shape, weaving in and out of the metal backing. A gas passageway **314** is coupled to the gas channels **310** to provide gases to the gas channels **310** during operation. The gas channels **310** and the gas passageway **314** are anodized to protect the gas channels **310** and gas passageway **314** from corrosive gases, as shown by the dotted lines **340** in FIG. **3**C, as the gas channels **310** and gas passageway **314** are in contact with process gases during operation. The gas channels **310** of the showerhead plate **128** are recessed from the metal backing, like shown in FIGS. **4**A-**4**B. As discussed above, the GDP **302** has a thickness in the y-direction of about **2.5** mm, which is significantly thinner than the conventional GDP **202** of FIG. **2**. Due to the thinness of the GDP **302**, the plurality of holes **360** in the GDP **302** may be mechanically drilled or laser drilled. Mechanically drilling the plurality of holes **360** is less expensive than laser drilling. Moreover, as the GDP **302** is much thinner, the holes **360** can be formed more rapidly, with less damage, thus further saving time and fabrication costs as compared to fabricating the much thicker, conventional GDP **202**.

[0041] The heat transfer plate **130** comprises cooling channels **318** disposed on a first side **321** of the showerhead assembly **120**, and a cap **319** is disposed between the first side **321** and the cooling channels **318**. The cooling channels **318** are grooves or slots disposed within the heat transfer plate **130** that have a cap **321** welded over the cooling channels **318** to allow a coolant or other heat transfer fluid to flow therethrough. The heat transfer plate **130** comprises a first side **322** and a second side **324** disposed opposite the first side **322**. The first side **322** has a larger outer diameter than the outer diameter of the second side **324**. One or more heaters **316** are disposed in one or more heater receiving channels **317** disposed on the second side **324** of the heat transfer plate **130**. The one or more heaters **316** are sealed in the heater receiving channels **317** with one or more plugs **323**, where the one or more plugs **323** are welded to the heat transfer plate **130**.

[0042] The first and second sides **322**, **324** are coupled together by a tapered sidewall **320**, where the tapered sidewall **320** extends from the first surface **322** near the lid **131** to the showerhead plate **128** (i.e., the tapered sidewall **320** is tapered from the first surface **322** to the second surface **324** in the -y direction). The tapered sidewall **320** of the heat transfer plate **130** is disposed to a tapered sidewall **332** of the showerhead plate **128**. The tapered sidewall **332** of the showerhead plate **128** extends from the second side **324** of the heat transfer plate **130** to a bottom surface **333** of the showerhead plate **128**. A pressure sensor **336** is disposed over the lid **131**, and a pressure sensing channel **338** runs down the sidewall **320**, like shown in FIG. **5**.

[0043] As shown in FIG. **3**B, the GDP **302** is bonded to the bottom surface **333** of the showerhead plate **128** with a thermally and electrically conductive adhesive **326**. The bottom surface **333** is bare, non-anodized metal so that an electrical connection can be made through the adhesive **326**. The adhesive **326** maintains a ground path between the GDP **302** and the heat transfer plate **130**. A portion of the adhesive **326** disposed on the GDP **302** extends over the anodized gas channels **310** to ensure the bare metal bottom surface **333** of the showerhead plate **128** is protect from the gases flowing through the paths **310**. The adhesive **326** is silicon based and comprises carbon and aluminum. The adhesive **326** may have a thickness of about 0.1 mm to about 0.2 mm.

[0044] Because the showerhead plate **128** and the GDP **302** are coupled together by the adhesive **326** and are distinct from the heat transfer plate **130** (i.e., not embedded within the heat transfer plate **130**, unlike a conventional showerhead assembly), the showerhead plate **128** and the GDP **302** can be easily replaced when needed.

[0045] FIG. **4**A illustrates a bottom view of the showerhead plate **128** of the showerhead assembly **120** of FIGS. **3**A-**3**C that includes two or more gas delivery zones, according to one embodiment. FIG. **4**B illustrates a bottom view of the showerhead plate **128** of the showerhead assembly **120** comprising three or more zones, according to another embodiment.

[0046] As shown in FIG. **4**A, the showerhead assembly **120** includes at least an outer zone **342** and an inner zone **344**. Both the outer and inner zones **342**, **344** comprise individual (e.g., separate) plenums connected to separate gas passageways **314**. The outer and inner zones **342**, **344** are fluidly isolated and separated by a ring formed from the adhesive **326** coupling the GDP **302** to the showerhead plate **128**, thus enabling the amount and/or type of gas delivered to each zone **342**, **344** to be separately controlled. Similarly, as shown in FIG. **4**B, the showerhead assembly **120** may comprise the outer zone **342**, an inner zone **344**, and a center zone **346** separated by a ring formed from the adhesive **326** coupling the GDP **302** to the showerhead plate **128**. In the showerhead assembly **120** of FIG. **4**B, the center zone **346** is extremely small, resulting in the gas flow to the center zone **346** being highly controllable. In the example depicted in FIG. **4**B, each of the zones **342-346** are concentric to a centerline of the showerhead assembly **120**.

[0047] In FIGS. **4**A and **4**B, the curved rectangular portions **352** illustrate raised mesas form on the bottom surface **333** of the showerhead plate **128**, and the recessed portions **354** defined between the curved rectangular portions **352**. The recessed portions **354** define the portion of the anodized gas channels **310** formed in the bottom surface **333** of the showerhead plate **128**. As such, the recessed portions **354** are anodized while the raised curved rectangular portions **352** are not (i.e., are bare aluminum). The adhesive **326** coupling the GDP **302** to the showerhead plate **128** is disposed on the bare raised curved rectangular portions **352** to provide good electrical conduction between the showerhead plate **128** and the GDP **302**. The plurality of holes **360** of the GDP **302** align with the recessed portions **354**.

[0048] FIG. **5** illustrates a pressure sensor **336** and a pressure sensing channel **338** of the showerhead assembly **120** of FIGS. **3**A-**3**C, according to one embodiment. As shown, a pressure sensing channel **338** is formed through the heat transfer plate **130**. The pressure sensing channel **338** extends between a top port **362** formed in the top surface **322** of the heat transfer plate **130** and a side port **364** formed in the sidewall **320** of the heat transfer plate **130**. The pressure sensing channel **338** may be a straight hole extending between the ports **362**, **364**, or have multiple connected sections. For example, in the embodiment depicted in FIG. **5**, the pressure sensing channel **338** includes a first section **366** extending form the top port **362** to a second section **368**. The second section **368** extends from the first section **362** to the side port **364**. In one example, the first section **366** has a centerline that is parallel to a centerline of the showerhead assembly **120** and/or the heat transfer plate **130**, while the second section **368** has a centerline that is perpendicular to the centerline of the first section **366**. The centerlines of one or both of the first and second sections **366**, **368** may have other orientation, or be combined with additional sections. As discussed above, the interior surfaces of the pressure sensing channel **338** are anodized to provide protection from process and other gases.

[0049] An adapter **358** disposed on the first surface **322** of the heat transfer plate **130** is coupled to the pressure sensor **336** via a conduit **370**. The pressure sensing channel **338** is coupled to the adapter **358** and has a first end **372** at the top of the conduit **370** near the pressure sensor **336**. The pressure sensing channel **338** runs down the sidewall **320** in a gap **374** formed between the heat transfer plate **130** and the isolator **122** or liner **348**, before continuing between the GDP **302** and a liner **348** of the showerhead assembly **120** to a second end **376** of the pressure sensing channel **338**. The second end **376** of the pressure sensing channel **338** is in fluid communication with the gap **374** to allow pressure to be read of the process region **112** by the pressure sensor **336** without having a port exposed on the bottom surface of GDP **302** and/or the showerhead plate **128**. The pressure of the gap **374** of the pressure sensing channel **338** is the same as the pressure of the processing chamber with a marginal time delay (i.e., less than one second), and thus, results in a substantially real-time response of the pressure of the chamber. While the first and second sections **366**, **368** of the pressure sensing channel **338** are shown as being L-shaped, the pressure sensing channel **338** may be drilled from the adapter **358** to the sidewall **320** at an angle (e.g., as one diagonal section).

[0050] In conventional showerhead assemblies, the pressure sensing channel extends from an upper port to a lower port, where the lower port faces a substrate within the processing chamber. Since the pressure sensing channel **338** of the showerhead assembly **120** does not require a lower port facing a substrate, the processing chamber has an overall thermal and processing uniformity. Furthermore, the pressure sensing channel **338** lacking a lower port prevents the pressure sensing channel **338** from being exposed to a plasma or gas during operation, as a lower port can cause non-uniformity in processing a substrate.

[0051] Therefore, due to the showerhead plate **128** and the GDP **302** being coupled together by the adhesive and being distinct from the heat transfer plate **130** (i.e., not embedded within the heat transfer plate **130** like a conventional showerhead assembly), the showerhead plate **128** and the GDP **302** can be easily replaced when needed. Moreover, the showerhead plate **128** can be anodized, and the thin GDP **302** may have mechanically drilled holes, thus resulting in reduced costs when compared to a conventional showerhead assembly.

[0052] While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A showerhead assembly, comprising: a heat transfer plate having a first side coupled to a second side by a sidewall, the heat transfer plate having cooling channels formed adjacent the first side of the heat transfer plate and a heater disposed in a heater receiving channel formed in the second side of the heat transfer plate; a showerhead plate having a first side coupled to the second side of the heat transfer plate; and a gas distribution plate bonded to a second side of the showerhead plate, the gas distribution plate having a first plurality of holes formed therethrough, first ends of the first plurality of holes open to the showerhead plate.
2. The showerhead assembly of claim 1, wherein the showerhead plate comprises a plurality of gas channels formed in a bottom surface of the showerhead plate, the plurality of gas channels being bounded on one side by the gas distribution plate.
3. The showerhead assembly of claim 2, wherein the plurality of gas channels are anodized.
4. The showerhead assembly of claim 2, wherein the showerhead plate is bonded to the gas distribution plate using an adhesive, the adhesive comprising Si, Al, and C.
5. The showerhead assembly of claim 1, further comprising a first gas passageway extending through the heat transfer plate and fluidly coupled to the plurality of gas channels formed in the showerhead plate.
6. The showerhead assembly of claim 1, wherein the first plurality of holes are formed in a first zone of the gas distribution plate, and wherein the gas distribution plate further comprises a second zone having a second plurality of holes formed therethrough, first ends of the second plurality of holes being open to the to the showerhead plate.
7. The showerhead assembly of claim 6, wherein the gas distribution plate further comprises a third zone having a third plurality of holes formed therethrough, first ends of the third plurality of holes being open to the to the showerhead plate.
8. The showerhead assembly of claim 1, further comprising a pressure measurement path formed through the heat transfer plate, the pressure measurement path having a first end formed in the first side of the heat transfer plate and a second end formed in the sidewall.
9. A showerhead assembly, comprising: a heat transfer plate having a first side coupled to a second side by a sidewall, the heat transfer plate having cooling channels formed adjacent the first side of the heat transfer plate and a heater disposed in a heater receiving channel formed in the second side of the heat transfer plate; a showerhead plate having a first side coupled to the second side the heat transfer plate, wherein the showerhead plate comprises a plurality of anodized gas channels in a second side of the showerhead plate; a first gas passageway extending through the heat transfer plate to the plurality of anodized gas channels of the showerhead plate; and a gas distribution plate bonded to a second side of the showerhead plate, the gas distribution plate having a first plurality of holes formed therethrough, first ends of the first plurality of holes open to the plurality of anodized gas channels formed in the showerhead plate.
10. The showerhead assembly of claim 9, wherein the plurality of anodized gas channels of the showerhead plate are partially bounded by the gas distribution plate.
11. The showerhead assembly of claim 9, wherein the showerhead plate is bonded to the gas distribution plate using a thermally and electrically conductive adhesive, the thermally and electrically conductive adhesive comprising Si, Al, and C.
12. The showerhead assembly of claim 9, wherein the first plurality of holes are formed in a first zone of the gas distribution plate, and wherein the gas distribution plate further comprises a second zone having a second plurality of holes formed therethrough, first ends of the second plurality of holes being open to the showerhead plate.
13. The showerhead assembly of claim 12, wherein the gas distribution plate further comprises a third zone having a third plurality of holes formed therethrough, first ends of the third plurality of

holes being open to the showerhead plate.
14. The showerhead assembly of claim 9, further comprising a pressure sensing channel formed through the heat transfer plate, the pressure sensing channel having a first port formed in the first side of the heat transfer plate and a second port formed in the sidewall.
15. A processing chamber comprising: a chamber body having an internal volume; a substrate support disposed in the internal volume; a lid disposed on the chamber body and enclosing the internal volume; a heat transfer plate having a first side coupled to a second side by a sidewall, the heat transfer plate having cooling channels formed adjacent the first side of the heat transfer plate and a heater disposed in a heater receiving channel formed in the second side of the heat transfer plate; a showerhead plate having a first side coupled to the second side the heat transfer plate, wherein the showerhead plate comprises a plurality of gas channels in a second side of the showerhead plate; and a gas distribution plate bonded to a second side of the showerhead plate, the gas distribution plate having a first plurality of holes formed therethrough, first ends of the first plurality of holes open to the plurality of gas channels formed in the showerhead plate.
16. The showerhead assembly of claim 15, further comprising a first gas passageway extending through the heat transfer plate to the showerhead plate, wherein the first gas passageway and the plurality of gas channels are anodized.
17. The showerhead assembly of claim 15, wherein the showerhead plate is bonded to the gas distribution plate using an adhesive, the adhesive comprising Si, Al, and C.
18. The showerhead assembly of claim 15, wherein the first plurality of holes are formed in a first zone of the gas distribution plate, and wherein the gas distribution plate further comprises a second zone having a second plurality of holes formed therethrough, first ends of the second plurality of holes being open to the showerhead plate.
19. The showerhead assembly of claim 18, wherein the gas distribution plate further comprises a third zone having a third plurality of holes formed therethrough, first ends of the third plurality of holes being open to the showerhead plate.
20. The showerhead assembly of claim 15, further comprising a pressure measurement path formed through the heat transfer plate, the pressure measurement path having a first end formed in the first side of the heat transfer plate and a second end formed in the sidewall.
